

CATEGORY		HIGH-END INTEL MOTHERBOARDS			
		Aopen	Gigabyte	Intel	MSI
Brand		Aopen	Gigabyte	Intel	MSI
Model		AX4C Max	GA-8KNXP	875PBZ	875P NEO
Chipset		Intel 875P	Intel 875P	Intel 875P	Intel 875P
FEATURES (40%)	Maximum CPU speed supported (GHz)	3.2	3.2	3.2	3.2
	CPU FSB and Hyperthreading support (MHz)	800 / ✓	800 / ✓	800 / ✓	800 / ✓
	Memory supported (MHz) / Memory slots / Max memory (GB)	DDR 400 / 4 / 4	DDR 400 / 6 / 4	DDR 400 / 4 / 4	DDR 400 / 4 / 4
	Form factor	ATX	ATX	ATX	ATX
	AGP slot / Onboard AGP / Type	✓ / X / 8X	✓ / X / 8x AGP PRO	✓ / X / 8X	✓ / X / 8X
	No of PCI slots	5	5	5	5
	Onboard six-channel audio	✓	✓	No sound	✓
	Onboard FireWire / Firewire-B	✓ / X	✓ / X	X / X	✓ / X
	No of USB ports / Rear / Ports via cable	8 / 6 / 2	8 / 4 / 4	8 / 6 / 2	8 / 6 / 2
	Ethernet / Gigabit Ethernet	X / ✓	X / ✓	X / ✓	X / ✓
	IDE RAID / Type supported RAID 0, 1, 0+1	X	✓ / All three modes	X	✓ / Two modes
	SATA connectors on motherboard / SATA RAID	4 / ✓	4 / ✓	4 / ✓	4 / ✓
	Core voltage / Frequency adjustment	✓ / ✓	✓ / ✓	X / X	✓ / ✓
	Overclocking and BIOS updating features / software	✓	✓	X	✓
	Component layout on motherboard (S05)	2.5	3.5	3.5	3
Package contents					
Manual / CDs		✓ / ✓	✓ / ✓	✓ / ✓	✓ / ✓
IDE, SATA and power cables provided		✓	✓	✓	✓
Other miscellaneous components offered		X	DPS and external SATA bracket	X	Sound, D-led and Firewire bracket
PERFORMANCE (60%)	Gaming Benchmarks: Serious Sam Second Encounter (fps)				
	Average fps	123.72	123.28	136.46	136.34
	UT2003 Flyby (fps)				
	Average fps	199.97	199.97	202.19	202.68
	Quake3 Arena (fps)				
	Average fps	342.27	341.53	408.67	393.90
	Real World benchmarks				
	ZDbench Content Creation 2004	27.7	27.7	28	28
	ZDbench Business Winstone 2004	22.7	22.8	23.3	23.1
	Video encoding (secs)	71	71	69	70
	Synthetic Benchmarks				
	3D Mark 2003 (patch 340)	5,588	5,577	5,608	5,723
	PC Mark 2004				
	CPU index	4,998	4,488	4,900	4,942
	Memory index	4,659	4,600	4,717	4,710
	Graphics	5,287	5,131	5,280	5,317
	Disk Index	0	2,901	2,937	2,932
	Total PC Marks	4,928	5,006	5,086	5,126
	Sisoft Sandra 2004 Professional: CPU sub-system				
	CPU Dhrystone / Whetstone	9,739 / 4,022	9,637 / 4,011	9,215 / 3,637	9,710 / 4,059
	Multimedia - CPU Integer / FPU SSE	24,603 / 35,349	24,586 / 35,691	24,419 / 35,458	24,837 / 35,790
	Memory sub-system				
	ALU / RAM	4,626	4,621	4,870	4,830
	FPU / RAM	4,633	4,622	4,799	4,838
	Disk drive				
	Drive Index MBps	24,232	24,579	24,568	24,544
OVERALL SCORE	Performance	45.19	44.98	45.97	45.99
	Features	27	31.4	13.4	29.4
	Grand Total (Performance + Features)	72.19	76.38	59.42	75.39
	VFM (Index)	4.55	5.09	5.19	5.01
	Price (Rs)	15,850	15,000	11,440	15,000
Grade		B-	B	A+	A

*Scale of 5

Conclusion

All these boards, which are based on the 875 chipset, performed brilliantly. What differentiates them from the 865s is the performance acceleration technology (PAT), i.e., the optimisation of memory pathway between the CPU and the system memory. This optimisation results in a reduction of latency by about 2 clocks—a gain that is clearly reflected by the scores.

The MSI 875P Neo should make an ideal choice for any enthusiast—it offers the best performance with features that

are second only to the Gigabyte. Neo's easy and vast overclocking options add up to give us the best performance winner. The Gigabyte and AOpen boards are equally great, but were completely outshone by the Intel and MSI boards.

Gigabyte's GA-8KNXP, is loaded to the hilt with equally good BIOS tweaks for overclocking. Intel's 875PBZ on the other hand is a fire cracker at performance, but loses out on BIOS overclocking tweaks. Considering the Intel's outstanding performance, legendary stability and unbeatable price, we had no doubts while awarding it our Best Value award.